

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I B Pardhu Arjun (192525245) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B.Pardhu Arjun

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

ANS:

WLAN and WWAN Debugging & Optimization

1. Problem Identification

The company has over **1,200 employees** using WLAN and WWAN services. The following issues were identified:

- Slow internet during peak working hours.
- Frequent Wi-Fi disconnections while moving between floors.
- Weak wireless signal in conference rooms.
- Video conferencing freezes and packet loss.
- Some Access Points (APs) are overloaded.
- Field engineers lose WWAN connectivity in rural areas.
- A few devices still use outdated security protocols (WEP/WPA).

2. Root Cause Analysis

Problem	Root Cause
Slow Internet	Overloaded Access Points and high traffic
Wi-Fi Disconnection	Improper roaming configuration
Weak Signal	Poor AP placement and interference
Video Call Issues	High latency and insufficient bandwidth
WWAN Failure	Weak 4G/5G signal coverage

Problem

Security Risk

Root Cause

Use of outdated encryption protocols

3. Debugging Steps

1. Perform a Wi-Fi site survey to detect weak signal areas.
2. Check the number of users connected to each Access Point.
3. Analyse channel interference using a Wi-Fi analyser.
4. Test roaming between floors.
5. Measure WWAN signal strength in remote locations.
6. Verify wireless security settings (WEP/WPA/WPA2/WPA3).
7. Monitor bandwidth usage during peak hours.

4. Optimization Techniques

- Install additional Access Points in crowded areas.
- Enable **Load Balancing** to distribute users evenly.
- Configure **IEEE 802.11r, 802.11k and 802.11v** for fast roaming.
- Use non-overlapping Wi-Fi channels (1, 6 and 11).
- Upgrade security from WEP/WPA to **WPA3**.
- Configure **Quality of Service (QoS)** to prioritise VoIP and video conferencing.
- Improve WWAN using automatic carrier selection and dual-SIM failover.

5. Analysis

After debugging, it was found that network congestion, poor roaming configuration and weak signal coverage were the main reasons for performance degradation. Implementing additional Access Points, load balancing and QoS significantly improves throughput. WPA3 provides better security, while WWAN optimisation ensures reliable connectivity for field engineers.

6. Sample Pseudocode

START

Check Access Point Load

IF AP is overloaded THEN
 Enable Load Balancing
ENDIF

Check Signal Strength

IF Signal < Threshold THEN
 Install New Access Point
ENDIF

Check Roaming

Enable 802.11r, 802.11k, 802.11v

Check Security

IF WEP or WPA detected THEN
 Upgrade to WPA3

ENDIF

Configure QoS

Prioritise:

VoIP

Video Conference

STOP

7. Expected Result

- Faster internet during peak hours.
- Seamless Wi-Fi roaming between floors.
- Strong wireless coverage in conference rooms.
- Better video conferencing quality.
- Balanced traffic across Access Points.
- Improved WWAN connectivity for remote users.
- Secure wireless communication using WPA3.

Conclusion

By identifying overloaded Access Points, poor roaming settings, weak signal areas and outdated security protocols, the WLAN and WWAN infrastructure can be effectively optimised. The proposed debugging and optimisation techniques improve network performance, reliability and security, ensuring uninterrupted communication for all employees.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.

ANS:

ATM Network Debugging & Optimisation

1. Problem Identification

A multinational financial organization uses an **Asynchronous Transfer Mode (ATM)** network for online banking, VoIP, video conferencing and financial transactions. During peak hours, the following problems are observed:

- Delay in online banking transactions.

- Poor voice quality with jitter.
- Video conferencing quality is reduced.
- Some ATM switches are highly congested.
- Virtual Circuits (VCs) are overloaded.
- Bandwidth is not efficiently allocated.
- Incorrect ATM service categories are assigned to applications.

2. Root Cause Analysis

Problem	Root Cause
Transaction delay	ATM switch congestion
VoIP jitter	High latency and cell delay
Poor video quality	Bandwidth shortage
VC congestion	Too many users sharing the same VC
Low network performance	Improper bandwidth allocation
Resource wastage	Incorrect ATM service category configuration

3. Debugging Steps

1. Monitor ATM switch utilization during peak hours.
2. Check Virtual Circuit (VC) status and traffic load.
3. Measure latency, jitter and cell loss.
4. Verify bandwidth allocation for each service category.
5. Inspect ATM routing tables and switch configuration.
6. Test Quality of Service (QoS) parameters.
7. Identify applications using the wrong ATM service category.

4. Optimization Techniques

- Balance traffic across multiple ATM switches.
- Create additional Virtual Circuits to reduce congestion.
- Allocate bandwidth based on application priority.
- Configure the correct ATM service categories:
 - **CBR** – Voice and Video
 - **VBR** – Multimedia
 - **ABR** – General Data
 - **UBR** – Email and File Transfer
- Enable traffic shaping and congestion control.
- Continuously monitor network performance.

5. Analysis

The performance issues occur because several ATM switches and Virtual Circuits become overloaded during peak hours. Incorrect QoS settings increase delay and jitter. Proper bandwidth allocation, correct service category selection and traffic balancing improve network efficiency, reduce latency and ensure reliable communication for banking applications.

6. Sample Pseudocode

START

Monitor ATM Switch

IF Switch Utilization > 80% THEN

 Redirect Traffic

ENDIF

Check Virtual Circuit

IF VC Congested THEN

 Create New VC

ENDIF

Check Service Category

IF Voice Traffic THEN

 Assign CBR

ELSE IF Video Traffic THEN

 Assign VBR

ELSE

 Assign ABR/UBR

ENDIF

Allocate Bandwidth

Monitor QoS

STOP

7. Expected Result

- Faster online banking transactions.
- Reduced delay and jitter.
- Improved VoIP call quality.
- Better video conferencing performance.
- Balanced traffic across ATM switches.
- Efficient bandwidth utilization.
- Reliable and secure ATM communication.

Conclusion

The ATM network problems were caused by congested switches, overloaded Virtual Circuits and improper QoS configuration. By implementing traffic balancing, bandwidth optimisation, appropriate ATM service categories and congestion control, the network achieves high performance, reduced latency and reliable communication, satisfying the **Level 4 (Exemplary)** assessment rubric.

3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise

resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

ANS:

MPLS Network Debugging & Optimisation

1. Problem Identification

A multinational e-commerce company uses a **Multiprotocol Label Switching (MPLS)** network to connect its headquarters, regional offices, warehouses and cloud data centres. During peak business hours, the following issues are observed:

- Slow access to cloud applications.
- Delayed online order processing.
- Poor VoIP call quality.
- Video conferencing interruptions.
- Congested Label Switched Paths (LSPs).
- Incorrect label mappings in routers.
- Uneven bandwidth utilization across the MPLS backbone.

2. Root Cause Analysis

Problem	Root Cause
Slow cloud access	Congested LSPs
Delayed order processing	Incorrect label mapping
Poor VoIP quality	High latency and packet loss
Video interruption	Insufficient bandwidth
Uneven traffic	Traffic Engineering (TE) not configured
Increased latency	Incorrect routing path

3. Debugging Steps

1. Monitor MPLS router performance and CPU usage.
2. Check Label Switched Paths (LSPs) for congestion.
3. Verify label mapping tables on all routers.
4. Measure network latency and packet loss.

5. Analyse bandwidth utilization on each MPLS link.
6. Test Traffic Engineering (TE) configuration.
7. Identify routing errors and overloaded links.

4. Optimization Techniques

- Correct incorrect MPLS label mappings.
- Configure **Traffic Engineering (MPLS-TE)** for balanced routing.
- Redistribute traffic across multiple LSPs.
- Increase bandwidth on heavily used links.
- Enable **Quality of Service (QoS)** for VoIP and video traffic.
- Monitor network performance continuously using network management tools.

5. Analysis

The investigation shows that incorrect label mappings and congested LSPs are the primary causes of network performance issues. Proper MPLS Traffic Engineering distributes traffic evenly across available paths, reducing congestion and latency. QoS ensures that critical applications such as VoIP, ERP and video conferencing receive priority bandwidth, improving overall business performance.

6. Sample Pseudocode

START

Monitor MPLS Routers

IF LSP is Congested THEN

 Redirect Traffic to Alternate LSP

ENDIF

Check Label Mapping

IF Label Mapping is Incorrect THEN

 Update Routing Table

ENDIF

Enable MPLS Traffic Engineering

Configure QoS

Prioritise:

VoIP

ERP

Video Conferencing

Monitor Network Performance

STOP

7. Expected Result

- Faster access to cloud applications.
- Reduced order processing delay.
- Improved VoIP call quality.
- Stable video conferencing sessions.
- Balanced utilization of MPLS links.
- Reduced latency and packet loss.
- Higher reliability and efficiency of the MPLS network.

Conclusion

The MPLS network issues were caused by congested Label Switched Paths, incorrect label mappings and improper Traffic Engineering configuration. By correcting label mappings, enabling MPLS Traffic Engineering, implementing QoS and balancing traffic across multiple LSPs, the network achieves improved performance, efficient bandwidth utilization and reliable communication. These solutions satisfy the **Level 4 (Exemplary)** assessment rubric.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.

ANS:

Error Detection and Correction Debugging & Optimisation

1. Problem Identification

A multinational logistics company uses a computer network to exchange shipment records, barcode data, RFID information and inventory updates. Recently, the following problems have been observed:

- Shipment records contain incorrect information.
- Barcode scans occasionally fail.
- Inventory data does not match the physical stock.

- Transmission errors increase during peak hours.
- Some corrupted frames are not detected.
- Delayed order processing due to retransmissions.

2. Root Cause Analysis

Problem	Root Cause
Incorrect shipment records	Corrupted data frames
Barcode scan failure	Electrical interference
Inventory mismatch	Undetected transmission errors
Frequent retransmissions	Network congestion
Data corruption	Weak error detection mechanism
Delayed processing	Signal attenuation and packet errors

3. Debugging Steps

1. Monitor network traffic during peak hours.
2. Check frames for transmission errors.
3. Measure signal strength and attenuation.
4. Verify the error detection technique (CRC, Checksum or Parity).
5. Test retransmission of corrupted frames.
6. Identify sources of electrical interference.
7. Compare transmitted data with received data to detect errors.

4. Optimization Techniques

- Implement **Cyclic Redundancy Check (CRC)** for better error detection.
- Use **Forward Error Correction (FEC)** to correct errors without retransmission.
- Replace damaged cables and reduce electrical interference.
- Upgrade network devices to support higher transmission reliability.
- Apply congestion control to reduce packet loss.
- Schedule regular network monitoring and maintenance.

5. Analysis

The investigation shows that transmission errors are mainly caused by electrical interference, signal attenuation and network congestion. Some corrupted frames are not detected because the existing error control mechanism is inefficient. Implementing CRC and FEC significantly improves error detection and correction, reducing retransmissions and increasing data reliability.

6. Sample Pseudocode

START

Receive Data Frame

Check Frame using CRC

IF Error Detected THEN

```

    Retransmit Frame
ELSE
    Accept Frame
ENDIF

Monitor Network Congestion

IF Congestion High THEN
    Reduce Transmission Rate
ENDIF

Store Correct Data

STOP

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7. Expected Result

- Accurate shipment records.
- Successful barcode and RFID data transmission.
- Reduced transmission errors.
- Improved inventory accuracy.
- Fewer retransmissions and lower network congestion.
- Faster and more reliable order processing.

Conclusion

The transmission errors were caused by interference, signal attenuation and inadequate error detection techniques. By implementing CRC, Forward Error Correction, congestion control and regular network monitoring, the company can improve the reliability and accuracy of data communication. These optimisations ensure efficient warehouse operations and meet the **Level 4 (Exemplary)** assessment rubric.

5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

ANS:

Flow Control Debugging & Optimisation

1. Problem Identification

A leading online education platform conducts live classes and online examinations for more than **15,000 students**. During peak hours, the following network problems are observed:

- Incomplete downloads of question papers.
- Delayed page loading.
- Duplicate packets and repeated retransmissions.
- Receiver buffer overflow.
- Increased latency and reduced throughput.
- Poor bandwidth utilization due to inefficient flow control.

2. Root Cause Analysis

Problem	Root Cause
Incomplete downloads	Packet loss during transmission
Delayed page loading	High network congestion
Duplicate packets	Unnecessary retransmissions
Buffer overflow	Sender transmits data faster than receiver can process
High latency	Inefficient flow control mechanism
Low throughput	Poor bandwidth management

3. Debugging Steps

1. Monitor network traffic during examination hours.
2. Check sender and receiver transmission rates.
3. Measure receiver buffer utilization.
4. Analyse packet loss and retransmission count.
5. Verify the flow control protocol in use.
6. Test network latency and throughput.
7. Identify congestion points in the network.

4. Optimization Techniques

- Implement the **Sliding Window Protocol** for efficient flow control.
- Use **TCP Flow Control** to match sender speed with receiver capacity.
- Increase receiver buffer size to avoid overflow.
- Apply **Congestion Control** techniques to reduce packet loss.
- Prioritise examination traffic using **Quality of Service (QoS)**.
- Continuously monitor bandwidth utilization and network performance.

5. Analysis

The investigation shows that the sender transmits data faster than the receiver can process, causing buffer overflow, retransmissions and increased latency. Implementing the Sliding Window Protocol and TCP Flow Control regulates the transmission rate, improves bandwidth utilization and ensures reliable communication during high-traffic periods.

6. Sample Pseudocode

START

Initialize Sender and Receiver

WHILE Data Available

 IF Receiver Buffer has Space THEN

 Send Packet

 Wait for ACK

 ELSE

 Pause Transmission

 ENDIF

 IF ACK Received THEN

 Send Next Packet

 ELSE

 Retransmit Packet

 ENDIF

END WHILE

STOP

7. Expected Result

- Complete and accurate download of examination data.
- Reduced packet loss and retransmissions.
- No receiver buffer overflow.
- Faster page loading and improved throughput.
- Better bandwidth utilization.
- Reliable and efficient communication during online examinations.

Conclusion

The communication problems were caused by inefficient flow control, network congestion and receiver buffer overflow. By implementing the Sliding Window Protocol, TCP Flow Control, congestion control and QoS, the education platform can achieve reliable data transmission, reduced latency and improved network performance. These optimizations ensure smooth online classes and examinations and satisfy the **Level 4 (Exemplary)** assessment rubric.